

TrendWatch AI — Social Media Intelligence Report

Use Case 2: Brand monitoring on Reddit · Generated 2026-06-09 03:48 UTC

Query: murder r/crime Limit: 138 Time: devvit_import

Data: Live Reddit (imported from Devvit app)

Report contents (assignment sections)

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1. Executive briefing

Source: rule_based_fallback · RAG + CoT LLM or rule-based fallback

Executive summary

TrendWatch AI analysed 138 Reddit posts. Negative sentiment is 98.6%. Crisis score is 60/100 (Medium).

Main concerns

- Negative sentiment is 99% of posts.
- Risk keywords appeared 6 times across posts.
- 92 high-engagement posts carry negative sentiment.

Crisis explanation

Negative sentiment is 99% of posts. Risk keywords appeared 6 times across posts. 92 high-engagement posts carry negative sentiment.

Recommended actions

- Monitor high-impact negative threads daily.
- Prepare a factual FAQ for top complaint topics.
- Escalate to customer support if refund/delay keywords spike.

Communication strategy

Address emerging topics: That / Murder / With, Murder / With / From, Murder / Murders / Case. Use empathetic, transparent tone; avoid dismissive responses.

Full LLM report (markdown)

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Recommended Actions

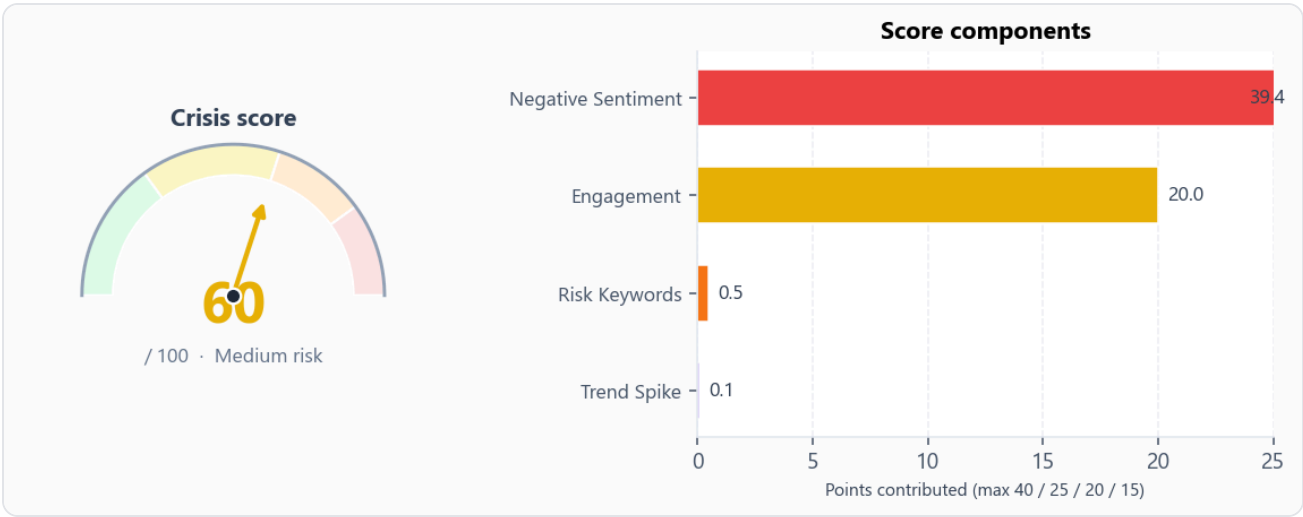
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Communication Strategy

Address emerging topics: That / Murder / With, Murder / With / From, Murder / Murders / Case. Use empathetic, transparent tone; avoid dismissive responses.

Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.

2. Crisis intelligence layer



60/100 — Medium risk

Weighted fusion of negative sentiment, risk keywords, high-engagement negative posts, and trend spikes.

Score components

Component	Points
negative_sentiment	39.4
risk_keywords	0.5
engagement	20.0
trend_spike	0.1

Reasons

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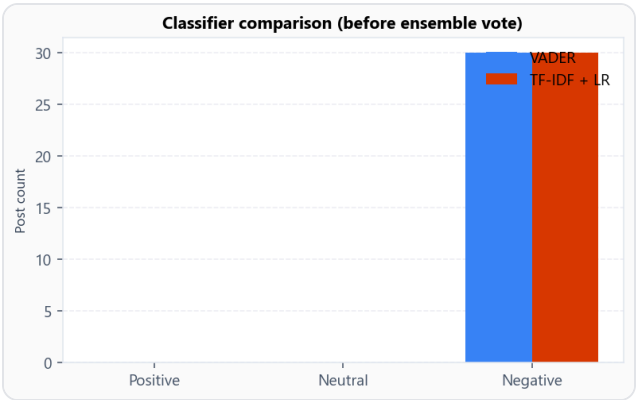
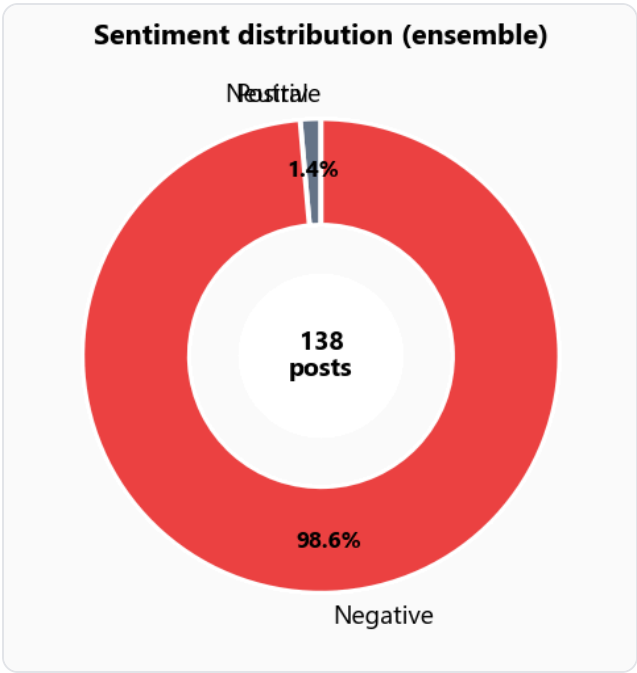
3. Overview & sentiment analysis

138
POSTS ANALYSED

0.0%
POSITIVE

1.4%
NEUTRAL

98.6%
NEGATIVE



Independent classifier outputs before ensemble fusion.

Final labels after VADER + LR ensemble majority vote.

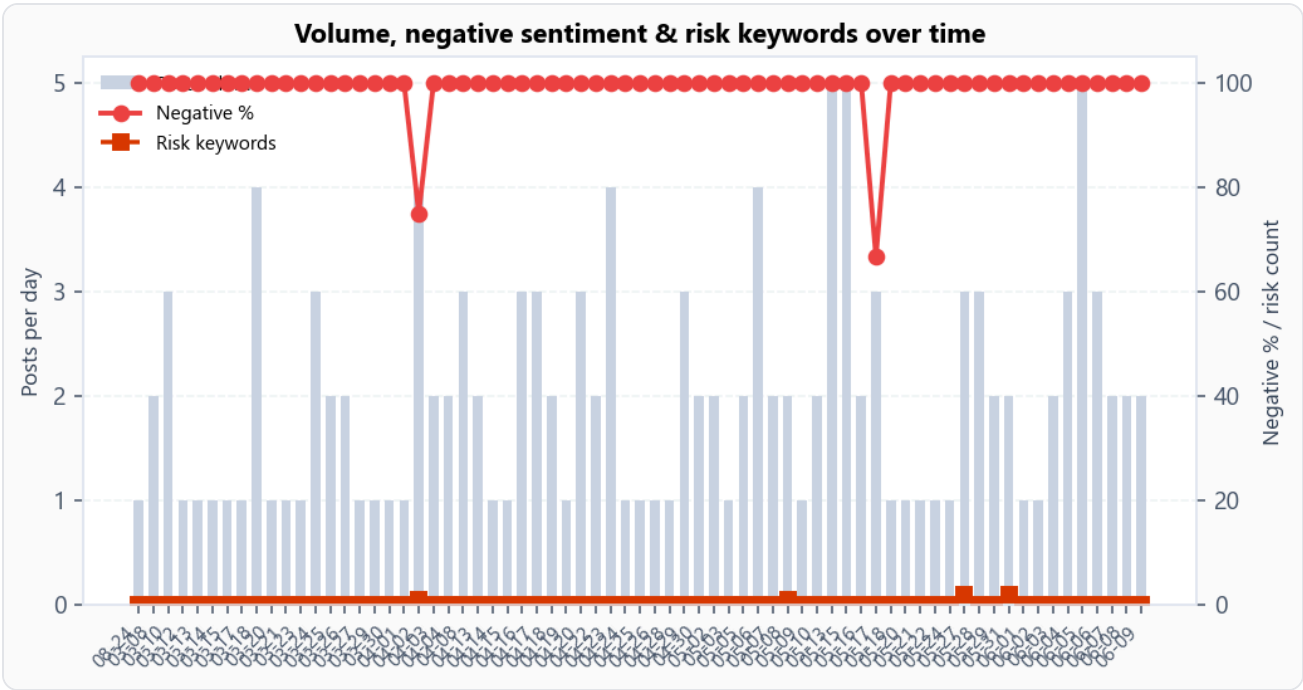
Sentiment distribution (VADER + LR ensemble)

Positive	0.0%
Neutral	1.4%
Negative	98.6%

Label	Count
positive	0
neutral	2
negative	136

Counts: positive=0, neutral=2, negative=136

4. Trend detection



Daily post volume (bars), negative sentiment % (line), and risk keyword counts (dashed).

Spike detected: No — No significant negative sentiment spike detected.

Average negative ratio: 99.2%

Date	Posts	Negative	Neg %	Risk keywords
2021-08-24	1	1	100.0%	0
2026-03-08	2	2	100.0%	0
2026-03-10	3	3	100.0%	0
2026-03-12	1	1	100.0%	0
2026-03-13	1	1	100.0%	0
2026-03-14	1	1	100.0%	0
2026-03-15	1	1	100.0%	0
2026-03-17	1	1	100.0%	0
2026-03-18	4	4	100.0%	0
2026-03-20	1	1	100.0%	0

Date	Posts	Negative	Neg %	Risk keywords
2026-03-21	1	1	100.0%	0
2026-03-23	1	1	100.0%	0
2026-03-24	3	3	100.0%	0
2026-03-25	2	2	100.0%	0
2026-03-26	2	2	100.0%	0
2026-03-27	1	1	100.0%	0
2026-03-29	1	1	100.0%	0
2026-03-30	1	1	100.0%	0
2026-04-01	1	1	100.0%	0
2026-04-02	4	3	75.0%	1
2026-04-03	2	2	100.0%	0
2026-04-04	2	2	100.0%	0
2026-04-08	3	3	100.0%	0
2026-04-13	2	2	100.0%	0
2026-04-14	1	1	100.0%	0
2026-04-15	1	1	100.0%	0
2026-04-16	3	3	100.0%	0
2026-04-17	3	3	100.0%	0
2026-04-18	2	2	100.0%	0
2026-04-19	1	1	100.0%	0
2026-04-20	3	3	100.0%	0
2026-04-22	2	2	100.0%	0
2026-04-23	4	4	100.0%	0
2026-04-24	1	1	100.0%	0
2026-04-25	1	1	100.0%	0

Date	Posts	Negative	Neg %	Risk keywords
2026-04-26	1	1	100.0%	0
2026-04-28	1	1	100.0%	0
2026-04-29	3	3	100.0%	0
2026-04-30	2	2	100.0%	0
2026-05-02	2	2	100.0%	0
2026-05-03	1	1	100.0%	0
2026-05-05	2	2	100.0%	0
2026-05-06	4	4	100.0%	0
2026-05-07	2	2	100.0%	0
2026-05-08	2	2	100.0%	1
2026-05-09	1	1	100.0%	0
2026-05-10	2	2	100.0%	0
2026-05-13	5	5	100.0%	0
2026-05-15	5	5	100.0%	0
2026-05-16	2	2	100.0%	0
2026-05-17	3	2	66.7%	0
2026-05-18	1	1	100.0%	0
2026-05-20	1	1	100.0%	0
2026-05-21	1	1	100.0%	0
2026-05-22	1	1	100.0%	0
2026-05-24	1	1	100.0%	0
2026-05-27	3	3	100.0%	2
2026-05-28	3	3	100.0%	0
2026-05-29	2	2	100.0%	0
2026-05-31	2	2	100.0%	2

Date	Posts	Negative	Neg %	Risk keywords
2026-06-01	1	1	100.0%	0
2026-06-02	1	1	100.0%	0
2026-06-03	2	2	100.0%	0
2026-06-04	3	3	100.0%	0
2026-06-05	5	5	100.0%	0
2026-06-06	3	3	100.0%	0
2026-06-07	2	2	100.0%	0
2026-06-08	2	2	100.0%	0
2026-06-09	2	2	100.0%	0

5. Topic clusters (embeddings + KMeans)

Topic 1: That / Murder / With

Size: 55 · Avg sentiment: **negative**

Keywords: that, murder, with, after, from, this

Representative: 5 Photos of Shamar Elkins: The Army Vet Who Murdered 8 Children in Shreveport After a Father-Daughter Date

Topic 2: Murder / With / From

Size: 33 · Avg sentiment: **negative**

Keywords: murder, with, from, that, charged, year

Representative: A man invaded my home, tortured, and tried to murder me in Atlanta

Topic 4: Murder / Murders / Case

Size: 22 · Avg sentiment: **negative**

Keywords: murder, murders, case, police, killer, with

Representative: Couple raped and murdered in case so disturbing that police needed therapy

Topic 0: Murder / That / This

Size: 16 · Avg sentiment: **negative**

Keywords: murder, that, this, trial, case, court

Representative: Court overturns Alex Murdaugh's murder convictions and orders new trial

Topic 3: Murder / Celeste / Rivas

Size: 12 · Avg sentiment: **negative**

Keywords: murder, celeste, rivas, after, hernandez, charge

Representative: D4vd Arrested for Murder After Celeste Rivas Hernandez, 15, Found Dead in His Tesla

6. Named entities (NER)

Entity	Mentions
two (CARDINAL)	10
2 (CARDINAL)	5
new york (GPE)	4
kouri (ORG)	4
fbi (ORG)	3
30 years (DATE)	3
decades (DATE)	3
15 year old (DATE)	3
india (GPE)	3
texas (GPE)	3
the next day (DATE)	3
14 year old (DATE)	3
arizona (GPE)	3
oregon (GPE)	2
henry nowak (PERSON)	2

7. Risk keywords & high-impact posts

Risk keyword frequency

Keyword	Count
leak	1
toxic	1
recall	1
broken	1
injury	1
terrible	1

Similar post pairs (cosine similarity)

Post A	Post B	Similarity
Two teenagers found guilty of murdering Kayden Moy, 16,	Two teenagers found guilty of murdering Kayden Moy at I	0.995
t3_1sqweeb	t3_1sqvuel	0.864
t3_1tc28qm	Court overturns Alex Murdaugh's murder convictions and	0.863
t3_1sqvuel	t3_1snnefb	0.827
t3_1sqweeb	t3_1snnefb	0.786

High-impact posts

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Couple raped and murdered in case so disturbing that police needed therapy	crime	491	66	negative	—	10.4
5 Photos of Shamar Elkins: The Army Vet Who Murdered 8 Children in Shreveport Af...	crime	337	91	negative	—	10.34
Alabama Man on Death Row for the Rape and	crime	334	43	negative	—	9.6

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Murder of 5-Year-Old Girl Wants Execut...						
A man invaded my home, tortured, and tried to murder me in Atlanta	crime	298	21	negative	recall, broken	8.79
Court overturns Alex Murdaugh's murder convictions and orders new trial	crime	180	35	negative	—	8.78

8. Post-level analysis (sample)

Showing up to 30 posts with VADER label, LR label, and extracted risk terms.

Title	Author	Ensemble	VADER	LR	Risk terms
Anna Kepner's Murder Case, FBI Recovered Teen's Smashed Cell Phone, Fa...	Bollywood_Shaadis	negative	negative	negative	—
The defense is now calling witnesses in the Karmelo Anthony murder tri...	southernemper0r	negative	negative	negative	—
Woman accused in teen girl's fatal fall from bridge denies murderous i...	LittleForm3711	negative	negative	negative	—
Oregon man, in court for allegedly killing 4 women, has just been arra...	RangerNew5346	negative	negative	negative	—
Man says he 'misplaced trust' in partner who 'sexually abused and murd...	IrishStarUS	negative	negative	negative	—
Police 'tried to smear Henry Nowak as aggressor' just three days after...	JohnKimble111	negative	negative	negative	—
Two teenagers found guilty of murdering Kayden Moy, 16, at Irvine beac...	TimesandSundayTimes	negative	negative	negative	—
Couple raped and murdered in case so disturbing that police needed the...	IrishStarUS	negative	negative	negative	—

Title	Author	Ensemble	VADER	LR	Risk terms
How a violent teenager on bail was left free to murder an innocent man	JohnKimble111	negative	negative	negative	—
'Why didn't you just leave?' Christine Banfield's family speaks after ...	BulkyHotel7719	negative	negative	negative	—
Pensacola man who confessed to killing father acquitted of his murder	Alternative-Room722	negative	negative	negative	—
Two teenagers found guilty of murdering Kayden Moy at Irvine beach	ScottishDailyRecord	negative	negative	negative	—
Saudi Arabian student charged with attempting to murder university sta...	JohnKimble111	negative	negative	negative	—
Melanie Hall's family launch fresh appeal for information 30 years aft...	prisongovernor	negative	negative	negative	—
Banfield remains convicted murderer, judge decides one day before sent...	BulkyHotel7719	negative	negative	negative	—
Accused Oregon serial killer charged with murdering 5 women: "My siste...	CBSnews	negative	negative	negative	—
A murder baffled investigators for decades. Then a Georgia man came fo...	cnn	negative	negative	negative	—
Twisted killer told 999 "he's dead" after murder of Good Samaritan in ...	ScottishDailyRecord	negative	negative	negative	—

Title	Author	Ensemble	VADER	LR	Risk terms
Lynette Hooker's Bahamas disappearance being investigated as possible ...	CBSnews	negative	negative	negative	—
Murdered student Henry Nowak told police 'I can't breathe' while handc...	Man_in_the_uk	negative	negative	negative	—
Key hearing for Charlie Kirk murder suspect will be open to reporters ...	CBSnews	negative	negative	negative	—
Man charged with murder in killings of 3 on Hawaii's Big Island	cnn	negative	negative	negative	—
I'm a psychology graduate & DA researcher. I read the text messages be...	Scary-Chemical-1370	negative	negative	negative	leak, toxic
Mother who murdered her baby in 'violent and brutal' attack jailed	TimesandSundayTimes	negative	negative	negative	—
California murder-suicide victims include 6-day-old baby—as police dis...	TheMirrorUS	negative	negative	negative	—
Louisville Reddit post helped police identify Michael Hristov as the s...	thecourierjournal	negative	negative	negative	—
88 year old man, beaten to death by 16 year old in Buffalo, NY. Senten...	Altgotcheck420	negative	negative	negative	—
A 15 year old boy brutally murdered in India, horror unfolds after see...	AccomplishedStore351	negative	negative	negative	—
Anna Kepner's Stepbrother Photographed for First Time as He Tries to S...	peoplemagazine	negative	negative	negative	—

Title	Author	Ensemble	VADER	LR	Risk terms
A man invaded my home, tortured, and tried to murder me in Atlanta	jena_raitor	negative	negative	negative	recall, broken

9. NLP techniques demonstrated (rubric)

Minimum **3 basic** + **3 advanced** techniques — implementation mapping for your written report.

Basic techniques

- Preprocessing (tokenize, stem, lemmatize, URL removal)
- Named entity recognition (spaCy / NLTK)
- Part-of-speech tagging
- TF-IDF + Logistic Regression classification
- Sentiment ensemble (VADER + LR majority vote)
- Embedding-based topic clustering (KMeans)
- Regex information extraction (hashtags, mentions, risk lexicon)

Advanced techniques

- Foundation LLM (OpenAI-compatible briefing)
- Retrieval-augmented generation (top-k posts)
- Prompt engineering (structured analyst template)
- Chain-of-thought prompting
- LLM-as-judge / automated evaluation
- Ensemble methods (sentiment + multi-signal crisis)
- Agentic ReAct-style orchestration
- Ethics & fairness disclaimers

10. Agent orchestration & evaluation

Agent status: completed

ReAct-style trace

Step	Thought	Action	Observation
1	Need Reddit posts about the brand/topic.	collect_posts	Starting data collection.
2	Collected 138 posts.	collect_posts	Received 138 posts.
3	Raw text must be normalized for NLP.	preprocess	Running tokenizer pipeline.
4	Preprocessing complete.	preprocess	138 posts prepared.
5	Extract hashtags, mentions, risk keywords.	extract	Regex IE.
6	Run NER and POS for entity-aware monitoring.	ner_pos	spaCy/NLTK.
7	Classify sentiment with VADER + LR ensemble.	sentiment	Ensemble vote.
8	Cluster topics with embeddings.	topics	KMeans clustering.
9	Compute temporal trends.	trends	Daily aggregation.
10	Score crisis risk from sentiment + engagement + trends.	crisis	Crisis formula.
11	Retrieve relevant posts for grounded LLM briefing (RAG).	rag	Query=murder
12	Generate executive briefing with CoT LLM.	briefing	LLM or fallback.

Automated evaluation

Sentiment agreement (VADER vs LR): 98.6% — VADER and LR agree on 136/138 posts (99%). Use manually labelled gold set for formal accuracy in your report.

Briefing judge score: 70/100 (error_fallback)

LLM judge unavailable; use manual rubric in report.

11. Ethics, limitations & references

--- *Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.*

- Public Reddit text only; no private user data.
- Crisis scores are decision-support estimates, not legal or PR advice.
- Sentiment models can misclassify sarcasm and domain-specific language.
- For quantitative evaluation on labelled data, extend `backend/evaluation.py`.
- Full NLP technique reference: `docs/NLP_TECHNIQUES.md`

TrendWatch AI · Devvit Web + Python FastAPI · [Raw JSON](#)